DSA ki Baithak

CODE BAITHAK

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Recap (Previous Lecture Connection):

* Conditional Statements
* Ternary Operator
* Loops
* break & continue Keywords
* Scope of a Variable

Teaser / Trailer :

```
*   *   *
  *   *
*   *
  *
*
```

```
* * * * *
  * * * *
    * * *
      * *
        *
      * *
    * * *
  * * * *
* * * * *
```

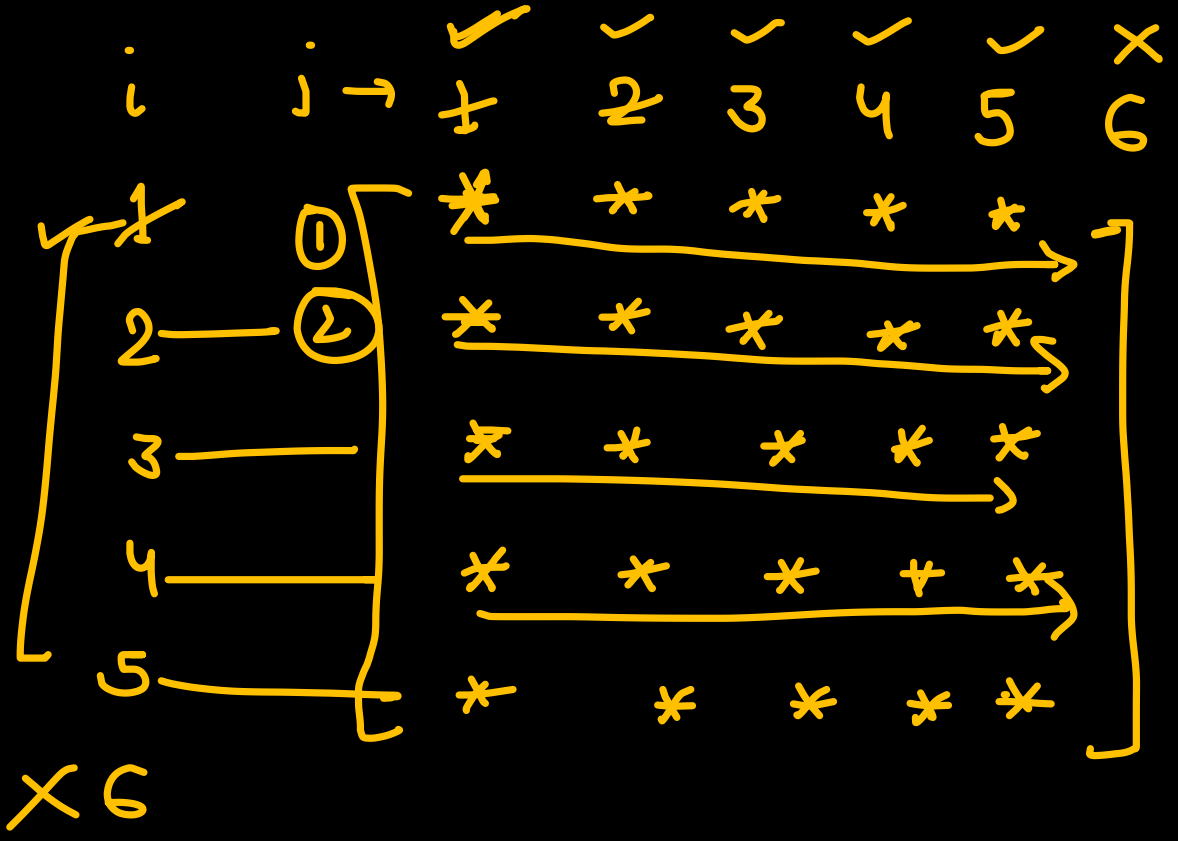
```
    *
  *   *
*       *
*       *
  *   *
    *
```

```
4           4
4 3       3 4
4 3 2     2 3 4
4 3 2 1   1 2 3 4
4 3 2 1 0 1 2 3 4
4 3 2 1   1 2 3 4
4 3 2     2 3 4
4 3       3 4
4           4
```

```
        1
      2 1   1 2
    3 2 1     1 2 3
4 3 2 1         1 2 3 4
    3 2 1     1 2 3
      2 1   1 2
        1
```

```
5 5 5 5 5 5 5 5 5
5 4 4 4 4 4 4 4 5
5 4 3 3 3 3 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 2 1 2 3 4 5
5 4 3 2 2 2 3 4 5
5 4 3 3 3 3 3 4 5
5 4 4 4 4 4 4 4 5
5 5 5 5 5 5 5 5 5
```

```
for (int i=1; i<=5; i++) {
    for (int j=1; j<=5; j++) {
        System.out.print(" * ");
    }
    System.out.println();
}
```



"Patterns Ka Gyaan...."

* Patterns always get printed rowwise (from left to right)

* Outer loop runs for the no. of rows

&

Inner loop runs for the no. of columns

Patterns Halwa Recipe

Steps -

1. First find out the total no. of rows in the pattern.
2. Find the tasks to be performed in each row of pattern.
3. Perform all the tasks of the current row.
4. Do preparation of the next row & change the line.

1.

$n = 5$

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

$n = 5$

1. Total rows → n
2. Star → n
3. —
4. —

2.

$n = 5$

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

1. Total rows $\longrightarrow n$
2. Digits $\longrightarrow n$
3.
4. —

char val = 'a' ;

3.

$n = 5$

a	a	a	a	a
b	b	b	b	b
c	c	c	c	c
d	d	d	d	d
e	e	e	e	e

1. Rows → n
2. Characters → n
3. —
4. ✓,

4.

$n = 5$

```
*
*  *
*  *  *
*  *  *  *
*  *  *  *  *
```

① rows → n

② stars → ~~x~~ → row

③ —

④ —

5.

$n = 5$

```
1
2 1
3 2 1
4 3 2 1
5 4 3 2 1
```

1. Rows → n
2. Digits → row
3. ___
4.

6.

$n = 5$

i \ j	1	2	3	4	5
1	1				
2	0	1			
3	1	0	1		
4	0	1	0	1	
5	1	0	1	0	1

even → 1

odd → 0

$i+j = 1+1 = 2$ → even

$= 2+1 = 3$ → odd → 0

$= 2+2 = 4$ → even → 1

7.

$n = 5$

```
* * * * *
* * * *
* * *
* *
*
```

1. Rows → n
2. Star → row
3. —
4. —

8.

$n = 5$

```
A B C D E
A B C D
A B C
A B
A
```

9.

$n = 5$

	col 1	2	3	4	5
row 1	*		*		*
2		*	-	*	
3	*		*		
4		*			
5	*				

row + col = 2+2 = 4 → even → *

row + col = 2+3 = 5 → odd → -

10.

$n = 5$

5	4	3	2	
5	4	3		1
5	4		2	1
5		3	2	1
	4	3	2	1